

Temporal Logic motion planning for multi-rotor UAVs in power tower inspection tasks

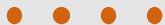
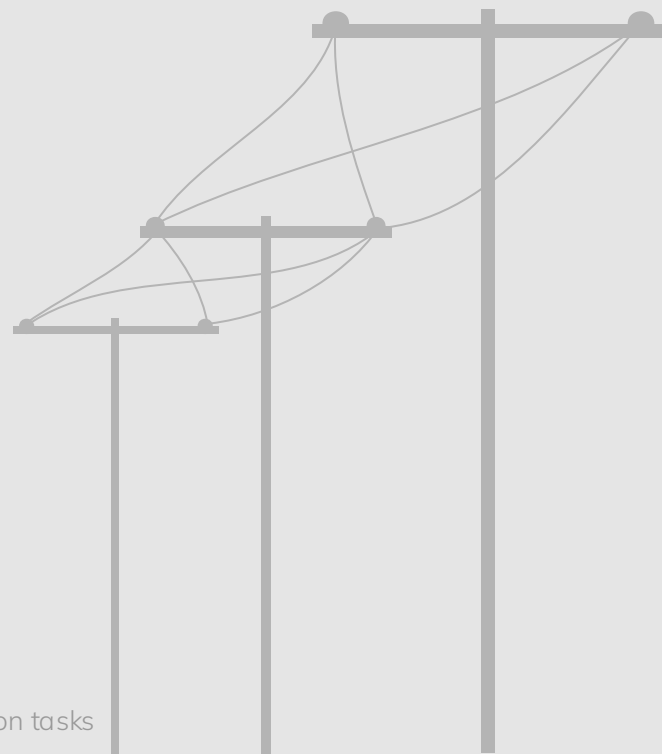
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Project overview

Context

- Global energy demand increase
- Major causes of a power outage is damage to electrical transmission lines
- Nowadays, approaches : manned helicopters equipped with multiple sensors

Solution

- Use of the multi-rotors Unmanned Aerial Vehicles (UAVs)
- Challenges due to electromagnetic interference and obstacles
- Signal Temporal Logic (STL) used to describe planning objectives



Project goals



Set up a **framework** using Signal Temporal Logic specifications to encode **inspection mission** for a fleet up to three quadrotors



- Identify **safety and mission requirements** for conducting inspection missions
- **Codify these requirements** using the Signal Temporal Logic formulation
- **Develop a mission** involving up to three UAVs, implementing basic safety features and inspection mission

Requirements identification

- Which elements ? **Insulators** and **earth rope**
- What type of power tower ? **220kV**
- Safety distance between drones and power tower ? Due to electromagnetic field, need to respect **8 m**

Source : Boukabou, I.; Kaabouch, N. Electric and Magnetic Fields Analysis of the Safety Distance for UAV Inspection around Extra-High Voltage Transmission Lines. *Drones* 2024, 8, 47.

Figure 6. The relationship between the maximum electric field strength of the surface of the UAV and the distance.

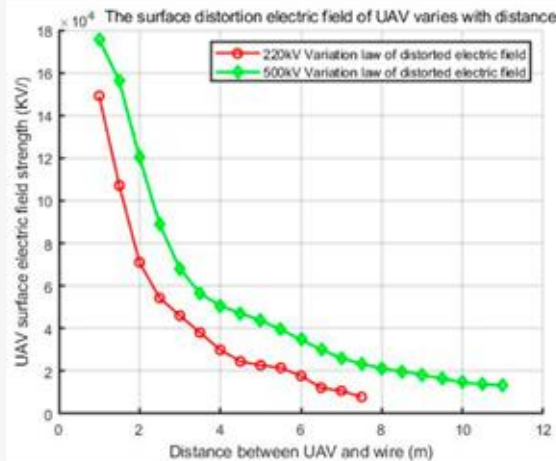
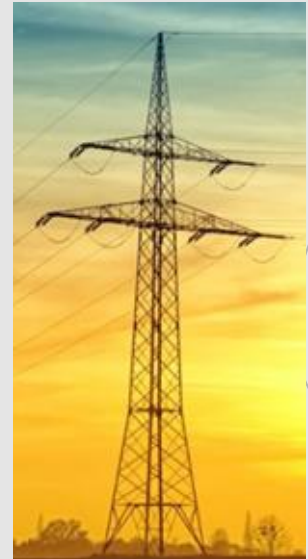
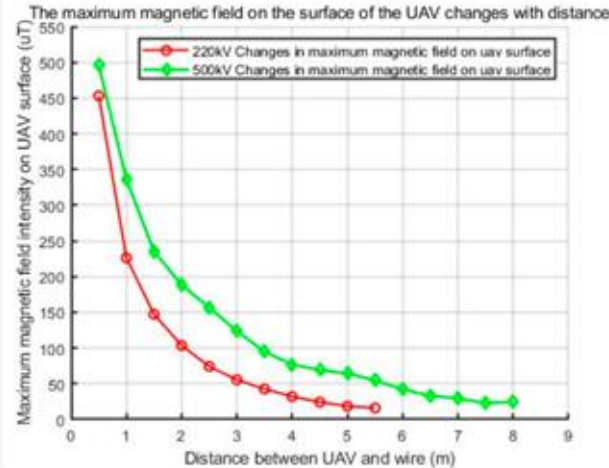


Figure 8. The variation of the surface magnetic field of the UAV with the distance.



Identification of the requirements

- Camera specifications

- Time to take an image : 3s
- Focal Length : 16mm
- Vertical number of pixels in the image : 3000
- Height of the image sensor : 4.6 mm

- UAV specifications

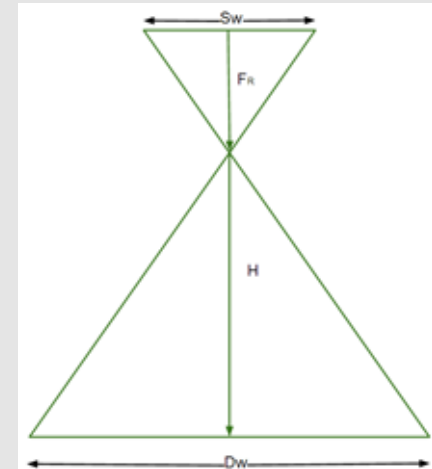
- Flight speed : 5 m/s
- Maximum flight time : 491 s (8 min)
- Maximum flight distance : 1856 m

Source : Baik, H., Valenzuela, J. Unmanned Aircraft System Path Planning for Visually Inspecting Electric Transmission Towers. J Intel Robot Syst 95, 1097–1111 (2019)

- Distance between drones and power tower to get good photo quality ?

- Size of insulator : 2 m
- Dw must be more than 2 m

Calculator		
Sw	4,6	= the sensor width of the camera (millimeters)
Fr	16	= the focal length of the camera (millimeters)
H	8	= the flight height (meters)
imW	3000	= the image width (pixels)
imH		= the image height (pixels)
GSD	0,08	= Ground Sampling Distance (centimeters/pixel)
Dw	2,30	= width of single image footprint on the ground (meters)
DH	0	= height of single image footprint on the ground (meters)



- Safety distance between drones ? 1 m



Source : <https://support.pix4d.com/hc/en-us/articles/202559809>

Problem formulation

$$\begin{aligned} \varphi = & \bigwedge_{d \in D} \Box_{[0, T_N]} ({}^d\varphi_{ws} \wedge {}^d\varphi_{obs} \wedge {}^d\varphi_{dis}) \wedge \\ & \bigwedge_{q=1} \Diamond_{[0, T_N - T_{ins}]} \bigvee_{d \in D} \Box_{[0, T_N]} {}^d\varphi_{tr, q} \wedge \\ & \bigwedge_{d \in D} \Box_{[1, T_N - 1]} ({}^d p(t_k) \models {}^d\varphi_{hm} \Rightarrow {}^d p(t_{k+1}) \models {}^d\varphi_{hm}) \end{aligned}$$

$${}^d\varphi_{ws} = \Lambda_{j=1}^3 p^{(j)} \in (\underline{p}_{ws}^{(j)}, \bar{p}_{ws}^{(j)}),$$

$${}^d\varphi_{obs} = \Lambda_{j=1}^3 \Lambda_{q=1}^{obs} p^{(j)} \notin (\underline{p}_{obs, q}^{(j)}, \bar{p}_{obs, q}^{(j)}),$$

$${}^d\varphi_{dis} = \Lambda_{\{d, m\} \in D, d \neq m} \| {}^d p - {}^m p \| \geq \Gamma_{dis},$$

$${}^d\varphi_{hm} = \Lambda_{j=1}^3 p^{(j)} \in (\underline{p}_{hm}^{(j)}, \bar{p}_{hm}^{(j)}),$$

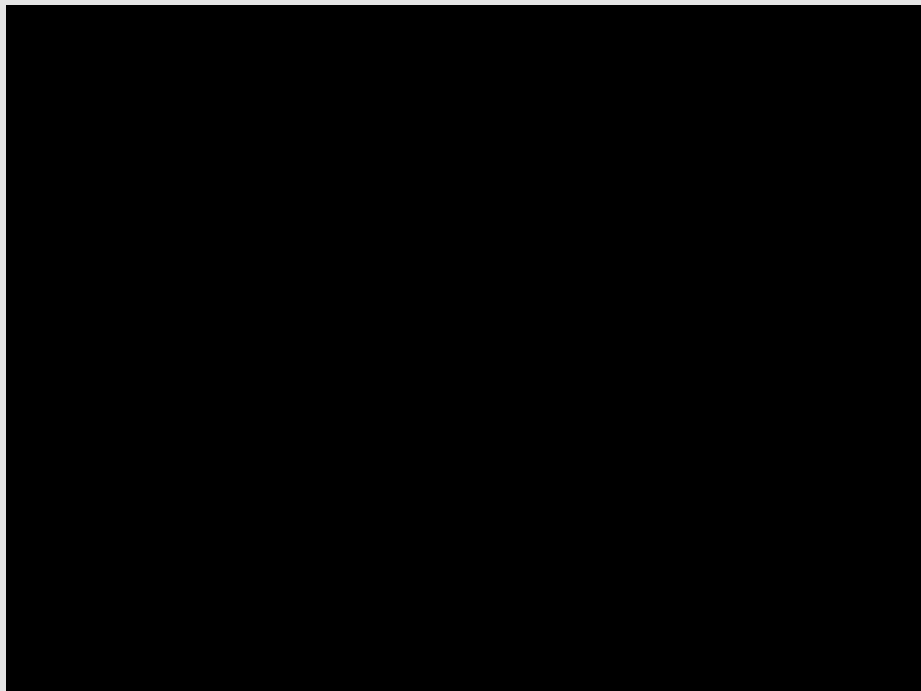
$${}^d\varphi_{tr, q} = \Lambda_{j=1}^3 {}^d p^{(j)} \in (\underline{p}_{tr, q}^{(j)}, \bar{p}_{tr, q}^{(j)}),$$

Source : A. Caballero and G. Silano, "A Signal Temporal Logic Motion Planner for Bird Diverter Installation Tasks with Multi-Robot Aerial Systems," *IEEE Access*, vol. 11, pp. 81361–81377, July, 2023.

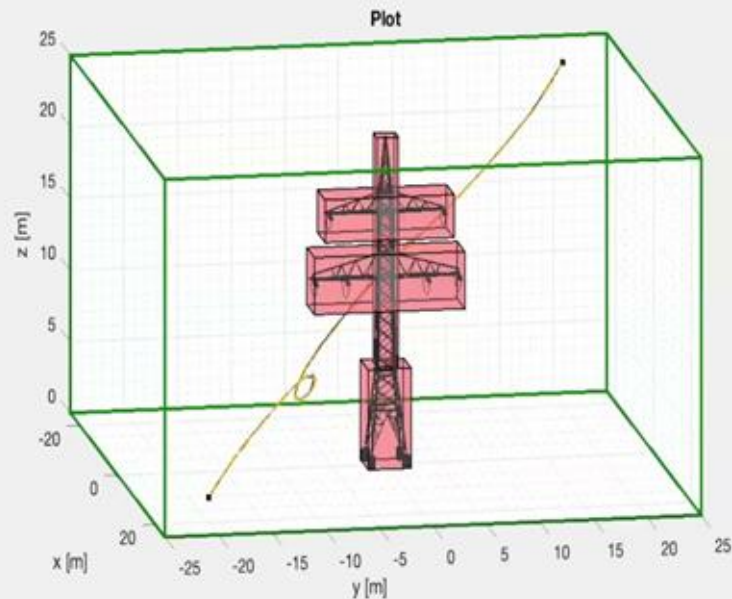


Simulation results 1

x2 Play speed

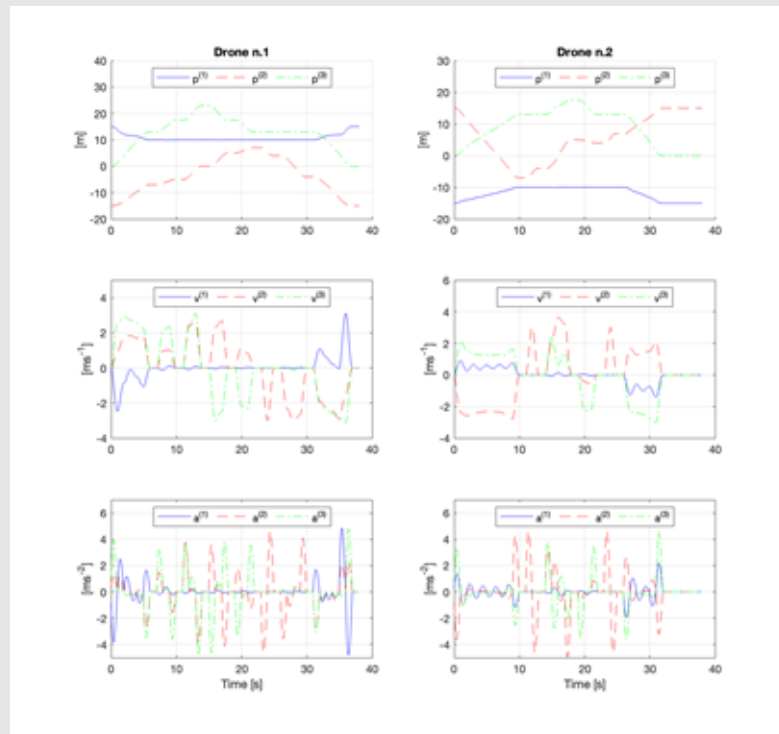
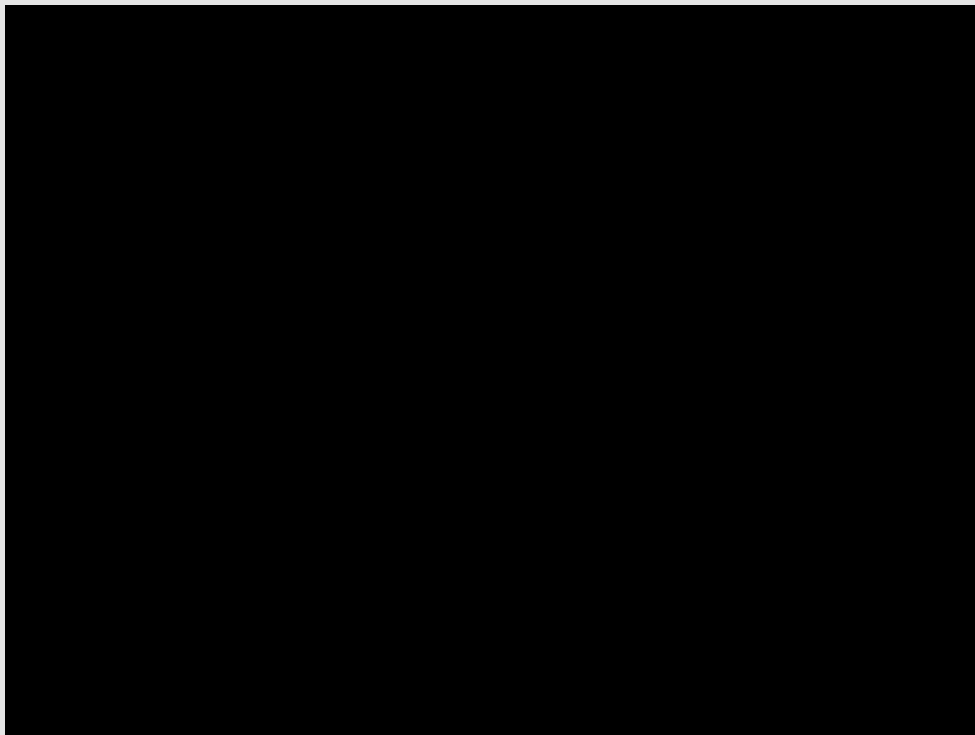


x2 Play speed



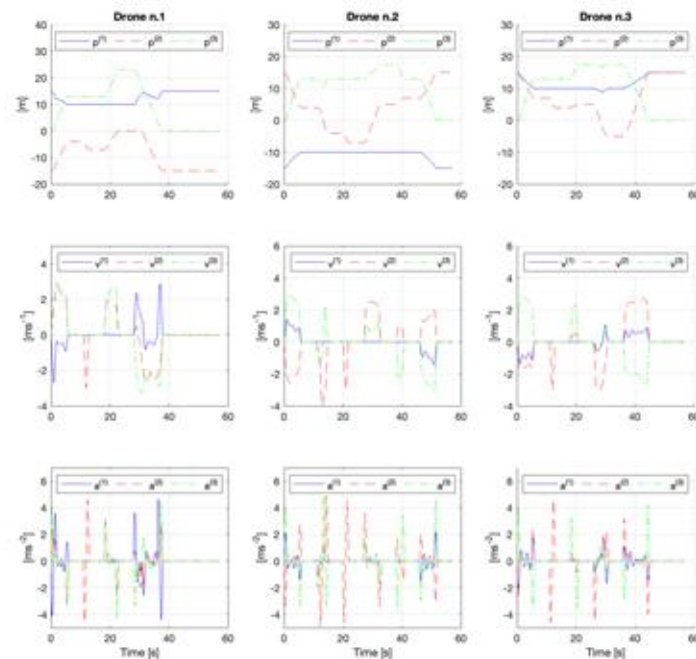
Simulation results 2

x2 Play speed



Simulation results 3

x2 Play speed





**Thank you for
your attention**

